

## Database Foundations

### 6-5: Defining Transaction Control Practices

#### Exercise 1: Controlling Transactions

##### Overview

In this practice you will control transactions using the following statements:

- COMMIT
- ROLLBACK
- SAVEPOINT

##### Assumptions

- Since TCL statements cannot be executed in Oracle Application Express (APEX) you will create the command language but not actually execute it.

##### Tasks

1. Suppose a table with this structure is created:

```
CREATE TABLE AD_STUDENT_TEST_DETAILS
(
  STUDENT_ID          NUMBER NOT NULL ,
  FIRST_NAME          VARCHAR2(50) ,
  STUDENT_REG_YEAR    DATE
) ;
```

Then the table is altered to add an email\_addr column:

```
ALTER TABLE AD_STUDENT_TEST_DETAILS ADD ( EMAIL_ADDR VARCHAR2(100)
UNIQUE );
```

After the ALTER a Savepoint is created called ALTER\_DONE.

A ROLLBACK is issued after the Savepoint ALTER\_DONE. Would the new email field still be there?

2. If an INSERT is done to add rows into the test table and a Savepoint is then created called INSERT\_DONE.

Then an UPDATE to a row in the test table is done and a Savepoint is created called UPDATE\_DONE.

Then a DELETE is executed to delete a row in the test table and a Savepoint is created called DELETE\_DONE. At this point what records would be in the table?

Then a ROLLBACK to Savepoint UPDATE\_DONE is issued. What changes would you notice with respect to the transactions and the records remaining in the table?

```
INSERT INTO AD_STUDENT_TEST_DETAILS VALUES(920, 'MAC', TO_DATE('01-JAN-2012','DD-MON-YYYY'),NULL);
INSERT INTO AD_STUDENT_TEST_DETAILS VALUES(940, 'RUTH', TO_DATE('01-SEP-2012','DD-MON-YYYY'),NULL);
INSERT INTO AD_STUDENT_TEST_DETAILS VALUES(950, 'ROBERT', TO_DATE('01-MAR-2012','DD-MON-YYYY'),NULL);
INSERT INTO AD_STUDENT_TEST_DETAILS VALUES(960, 'JEANNE', TO_DATE('01-MAR-2012','DD-MON-YYYY'),NULL);

SAVEPOINT CREATE_DONE;

UPDATE AD_STUDENT_TEST_DETAILS
SET EMAIL_ADDR = 'Mac@abc.com'
WHERE STUDENT_ID = 940;

SAVEPOINT UPDATE_DONE;

DELETE FROM AD_STUDENT_TEST_DETAILS WHERE STUDENT_ID = 950;

SAVEPOINT DELETE_DONE;

ROLLBACK TO UPDATE_DONE;
```